

Ganesh Gupta

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EXPERIENCE

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Designed and implemented a **Neo4j knowledge graph** modeling domain entities and relationships, serving as a reliable **context** source for both application and **AI agent** queries in production.
- Owned **applied AI/ML** infrastructure end-to-end, cutting **p99 latency 420ms to 140ms** across **50K+ daily requests** on an async, autoscaled architecture.
- Improved retrieval recall **61% to 89%** on a 2K-query eval set by rebuilding the pipeline with **hybrid dense-sparse retrieval** and re-ranking.

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Implemented **GraphRAG** (knowledge-graph-augmented retrieval) in production, improving grounding accuracy over standard vector retrieval for complex, multi-hop queries.
- Designed, scaled, and maintained **large-scale infrastructure**: a pluggable **distributed** executor routing workloads across **Docker** and **AWS Lambda** with an **S3-backed storage** layer.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Architected and operated **25+ production microservices** in **Java** and **C++** for **1M+ users** on **distributed systems**, sustaining **99.9% uptime** over 4 years.
- Standardized failure-handling contracts across 25+ services, reducing production escalations **60%** in one quarter via consistent retry, timeout, and propagation semantics.

PROJECTS

Distributed KV Store

github.com/ganeshhgupta/distributed-kv-store

- Built a **Raft-based distributed KV store** supporting concurrent writes, **linearizable reads**, failure recovery, and **log compaction**; companion VPC control plane simulator modeling subnet routing.

LangFetch

tinyurl.com/LangFetch

- LangGraph** agent with **Pinecone** vector retrieval and hybrid re-ranking; decomposes natural-language questions into dependency-ordered **SQL** subqueries via **FastAPI**.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Production evaluation platform with rubric-based scoring, hallucination detection, and drift tracking across model versions; **FastAPI** backend, **PostgreSQL**.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Java, Python, C++, TypeScript, SQL

Knowledge & Context Infra: Neo4j, Knowledge Graphs, GraphRAG, Entity-Relation Modeling, Graph-Based Retrieval

Distributed Systems: Large-Scale Infrastructure, Consensus Protocols, Compute & Storage, Fault Tolerance

Applied AI/ML: RAG, Vector Search, Model Evaluation, Agentic Workflows

Fundamentals: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, HTML, CSS

Cloud: AWS, GCP, Docker, Kubernetes, PostgreSQL